Quantum-to-Newton results: literature and Lean verification targets

Result and verification specification, 2026-10-27. The recent 0+1 construction yields several precise propositions worth formal checking, but their being new to this repository is not evidence of publication priority. The strongest distinctive candidates are the specified $C^{1,1}$ product/packet package and its sharp-rate example, the explicitly rescaled wall stresses, and the Gaussian preparation test at a bounded C^1 cusp. Each has substantial close prior art. The universal constructive-lifting conjecture and an independent positive-action necessity theorem remain **open**.

The targets below are exact mathematical specifications, **not Lean declarations that have been compiled**. Their written proofs are in the linked notes and are unrefereed; the $C^{1,1}$ lift received the internal Sol check recorded there. The targets state real quantifiers and boundary/domain conventions so a later Lean development can translate them without strengthening conclusions silently. A conditional theorem whose hypotheses include the conclusion, a placeholder quantum theory, or an unchecked imported axiom does not verify the corresponding concrete target.

1. What is already in the literature

Repository proposition	Closest checked precedent	Proper status
Coherent packets follow classical flow; smooth-observable semiclassics	Hepp (1974), abstract and theorem statements in the maintained note	Established mechanism, not a new proof of classical mechanics as such.
Uniform observable time-splitting estimates	Borns–Weil–Fang, Theorem 2 and §3.2 (passages read); Fang–Qu, Theorem 1 (statement read)	Prior uniform-in-$\hbar$ observable estimates. The former uses smooth bounded symbol classes; the latter treats periodic, spatially discretized polynomial observables and higher-order splittings. Neither passage alone establishes the exact $C^{1,1}$, arbitrary-partition, coherent-packet package below.
Reflected Gaussian and wall impulse	Belloni–Doncheski–Robinett (2004), equations (8)–(11), (16)–(19) (passages read); De Vincenzo (2013), equations (38)–(39) (passage read)	The odd-image Gaussian and collision-force analysis already exist. The particular measure limit, scaled profile and Neumann comparison in Target D are precise specializations whose exact priority remains unverified.

Repository proposition	Closest checked precedent	Proper status
Preparation-dependent singular-flow selection	Athanassoulis–Paul, Theorem 2.1 and Lemma 5.3 (statements and relevant passages read) include cutoff $-\|x\|^{1+\theta}$, including $\theta = 1/2$; Chabu, Theorem 1.12 and §3 (statements and passages read) gives identical incoming and different outgoing limits for $-\|x\|$.	Neither selection failure nor this power-law cusp is new as such. Target E specifies pure Gaussian preparations, a common incoming trajectory and their explicit mixtures for the bounded C^1 cusp. In Chabu's theorem the two profiles may also be chosen equal, so equality of initial position densities is not by itself a new general phenomenon. Exact coverage of Target E by prior results has not been settled.
Tangent groupoid, its continuous field and strict quantization	Cariñena et al. (1998/99), including Rivero, and Landsman–Ramazan (2000) (abstracts and groupoid passages read)	Established background; Target B is a particular state-level estimate on that field, not the invention of the groupoid or of strict quantization.
Normalized Euclidean tower	Brownian-bridge/Feynman–Kac and martingale constructions; Dimock I and II (selected sections in the guide)	Target G packages standard ingredients in a 0+1 illustration. Dimock's papers treat three-dimensional scalar ϕ^4, not this quantum-mechanical theorem or a four-dimensional gauge construction.
Relativistic one-particle limit	Carmona–Masters–Simon and Hiroshima–Ichinose–Lőrinczi (abstracts read)	Known relativistic Schrödinger and subordinate-process framework. Target F is a particular joint estimate; no literature priority has been established.

The table is a bounded search, not an exhaustive novelty determination. The primary proofs and their retained limitations remain in the Lipschitz lift, smooth construction, groupoid note, wall note, cusp note, and relativistic note.

2. Common definitions for Targets A–C

Use $\hbar > 0$, $m, \omega > 0$, $\mathcal{H} = L^2(\mathbb{R}, dq)$, $Qf(q) = qf(q)$, $P_\hbar = -i\hbar\partial_q$, and $N_\hbar = 1 + Q^2 + P_\hbar^2$ as the closure of the oscillator operator on Schwartz space, with $D(N_\hbar) = H^2(\mathbb{R}) \cap D(Q^2)$. Put

$$\|(q,p)\|_\omega = |q| + |p|/(m\omega), \quad K_s = e^{-isV(Q)/\hbar}, \quad D_s = e^{-isP_\hbar^2/(2m\hbar)}, \quad S_s = D_s K_s.$$

For a finite positive partition $\pi = (0 = t_0 < \cdots < t_n = T)$, write $s_j = t_{j+1} - t_j$, $|\pi| = \max_j s_j$, and $S_{\hbar,\pi}(t_j) = S_{s_{j-1}} \cdots S_{s_0}$, with the empty product at $j = 0$. The initial normalized Gaussian

is

$$g_\hbar(q) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \exp\left[-\frac{m\omega(q-q_0)^2}{2\hbar} + \frac{ip_0(q-q_0)}{\hbar}\right]. \tag{1}$$

The classical kick–drift vertices, starting at $z_0 = (q_0, p_0)$, obey

$$p_{j+1}^{\mathrm{cl}} = p_j^{\mathrm{cl}} - s_j V'(q_j^{\mathrm{cl}}), \qquad q_{j+1}^{\mathrm{cl}} = q_j^{\mathrm{cl}} + s_j p_{j+1}^{\mathrm{cl}}/m. \tag{2}$$

When used below, $z_{\hbar,\pi}(t)$ is the piecewise-linear interpolation of the **quantum means at the vertices**; it is not asserted to be the expectation of a physical intermediate state.

For the Weyl targets, use the Fourier convention

$$\widehat{b}(\xi,\eta) = \int_{\mathbb{R}^2} e^{-i(\xi q+\eta p)} b(q,p)\, dq\, dp, \quad \mathrm{Op}_\hbar^W(b) = \frac{1}{(2\pi)^2}\int \widehat{b}(\xi,\eta) e^{i(\xi Q+\eta P_\hbar)} d\xi d\eta. \tag{3}$$

This fixes the signs and factors needed by a formalization.

3. Target A: Lipschitz certificate, quantum product, and packet

Quantifiers and assumptions. Let $V \in C^1(\mathbb{R};\mathbb{R})$ and $L \geq 0$ satisfy $|V'(x) - V'(y)| \leq L|x-y|$ for every x, y. There is no lower bound assumption on V. For every $m, \omega, \hbar > 0$, $T > 0$ and $z_0 \in \mathbb{R}^2$, the following are to be proved.

1. The operator $H_\hbar = P_\hbar^2/(2m) + V(Q)$ with initial domain $\mathcal{S}(\mathbb{R})$ is essentially self-adjoint. Its closure generates $U_\hbar(t) = e^{-it\overline{H}_\hbar/\hbar}$, and this group preserves $D(N_\hbar)$ and is strongly continuous there in the $N_\hbar$ graph norm. On Schwartz space $[Q, P_\hbar] = i\hbar I$; for all $u, v \in \mathbb{R}$ the bounded Weyl operators satisfy $e^{iuQ}e^{ivP_\hbar} = e^{-i\hbar uv}e^{ivP_\hbar}e^{iuQ}$.
2. There exists $0 < C_{\hbar,T} < \infty$, depending on the supplied model and fixed $\hbar, T$, such that **every** positive partition satisfies

$$\|S_{\hbar,\pi}(T)f - U_\hbar(T)f\|_2 \leq C_{\hbar,T}|\pi|\, \|N_\hbar f\|_2 \quad (f \in D(N_\hbar)). \tag{A1}$$

   Consequently the product converges strongly on all of $\mathcal{H}$ as $|\pi| \downarrow 0$ at each fixed $\hbar$. For a general $\psi \in \mathcal{H}$ one precise modulus is the infimum over $f \in D(N_\hbar)$ of $2\|\psi - f\|_2 + C_{\hbar,T}\delta\|N_\hbar f\|_2$. The constant in (A1) is **not** asserted uniform in $\hbar$.
3. Let $z_j^\hbar = (\langle Q\rangle_j, \langle P_\hbar\rangle_j)$ in the state $S_{\hbar,\pi}(t_j)g_\hbar$, and let $W_j = \mathrm{sd}_j\, Q + \mathrm{sd}_j\, P_\hbar/(m\omega)$. Set $C = \sqrt{2/(m\omega)}$ and $\beta = \omega + L/(m\omega)$. Then for every vertex,

$$W_j \leq C\sqrt{\hbar}\, e^{\beta t_j}, \qquad \|z_j^\hbar - z_j^{\mathrm{cl}}\|_\omega \leq \frac{LC}{m\omega}\, t_j e^{\beta t_j}\sqrt{\hbar}. \tag{A2}$$

   The classical Lipschitz flow $\Phi_t(z_0)$ exists uniquely, and there is a finite classical constant R_T^{cl} such that $\sup_{0\leq t\leq T}\|z_\pi^{\mathrm{cl}}(t) - \Phi_t(z_0)\|_\omega \leq R_T^{\mathrm{cl}}|\pi|$. Hence the vertices converge to the flow for **every** joint limit $\hbar, |\pi| \downarrow 0$.

For the conjecture's bounded instance, the input is the *global* action/force and the classical polygon certificate. (A1) is proved from that model data plus the operator toolkit; it is not a consequence of the numerical error constant of a single orbit. Formalizing (A1) while assuming its self-adjointness or graph bounds should be reported as a conditional sublemma, not as the whole lift.

4. Target B: the smooth $O(\hbar + |\pi|)$ state estimate

Assume in addition that $V \geq 0$, $V \in C^\infty$ and all its derivatives of order at least two are bounded. Fix $h_0, T > 0$. Let $0 < \hbar \leq h_0$ and let π be any positive partition of $[0, T]$. In the notation above, write

$$\begin{aligned} a &= \max\left(\omega, \frac{L}{m\omega}\right), \qquad G_T = (e^{aT} - 1)/a, \\ b_0 &= \frac{|V'(0)| + LC\sqrt{h_0}e^{\beta T}}{m\omega}, \quad R_T = e^{\beta T}(\|z_0\|_\omega + Tb_0), \\ F_T &= |V'(0)| + L(R_T + C\sqrt{h_0}e^{\beta T}), \quad C_\delta = (F_T + LR_T)/m, \quad C_h = \frac{\|V'''\|_\infty C^2 e^{2\beta T}}{2m\omega}. \end{aligned} \tag{B0}$$

The specification is

$$\sup_{0 \leq t \leq T} \|z_{\hbar,\pi}(t) - \Phi_t(z_0)\|_\omega \leq G_T(C_\delta|\pi| + C_h\hbar). \tag{B1}$$

The centering step responsible for the better rate is the separate lemma

$$|\langle V'(Q)\rangle - V'(\langle Q\rangle)| \leq \tfrac{1}{2}\|V'''\|_\infty \operatorname{Var}(Q). \tag{B2}$$

It requires a normalized state in the needed second-moment domain. The first derivative term cancels; the width itself remains $O(\sqrt{\hbar})$.

For every $b \in C_c^\infty(\mathbb{R}^2)$ set

$$I_b = (2\pi)^{-2} \int |\widehat{b}|(|\xi| + m\omega|\eta|), \quad J_b = \frac{1}{2(2\pi)^2} \int |\widehat{b}|(|\xi| + m\omega|\eta|)^2. \tag{B3}$$

At every partition vertex the desired **concrete** result is

$$\left|\langle g_\hbar, S_{\hbar,\pi}(t_j)^* \operatorname{Op}_\hbar^W(b) S_{\hbar,\pi}(t_j) g_\hbar\rangle - b(\Phi_{t_j}(z_0))\right| \leq J_b C^2 e^{2\beta T}\hbar + I_b G_T(C_\delta|\pi| + C_h\hbar). \tag{B4}$$

It follows for any joint schedule $\hbar, |\pi| \to 0$. An additional continuous-field corollary takes $\hbar = \epsilon S_*$, $S_* > 0$, and extends the limit to continuous sections of the established tangent groupoid field: for every continuous section A, every family of partitions with $\epsilon, |\pi_\epsilon| \to 0$, and vertices $t_{j(\epsilon)} \to t \in [0, T]$, the corresponding vector-state value on $A(\epsilon)$ converges to $A(0)(\Phi_t(z_0))$. This corollary requires the continuous-field theorem and the identification of (3) with the chosen quantization section; (B4) alone does not construct that C^*-field.

5. Target C: sharpness of the square-root rate

Set $m = \omega = 1$, $V(q) = q|q|/2$, $q_0 = p_0 = 0$, and let $g_\hbar$ be (1). This potential is $C^{1,1}$ and **unbounded below**. The classical trajectory is identically $(0, 0)$. Define

$$f(t) = \langle U_1(t)g_1, P_1 U_1(t) g_1\rangle.$$

Prove, for every $\hbar > 0$ and real t,

$$\langle U_\hbar(t)g_\hbar, P_\hbar U_\hbar(t) g_\hbar\rangle = \sqrt{\hbar}\, f(t), \qquad f'(0) = -1/\sqrt{\pi}. \tag{C1}$$

In particular, there exists $t_0 > 0$ such that for all $0 < t \leq t_0$ and every $\hbar > 0$, the absolute momentum error is at least $t\sqrt{\hbar}/(2\sqrt{\pi})$. This refutes a uniform $O(\hbar)$ mean bound over the full class in Target A. It does not by itself establish sharpness under an additional lower-bound assumption on V.

6. Target D: Dirichlet and Neumann wall stresses

Let $m, \omega, q_0 > 0$, $p_0 < 0$, $v = p_0/m$, $t_* = -q_0/v$, $A(t) = 1 + \omega^2 t^2$, and $\Lambda = |v|\sqrt{m\omega/A(t_*)}$. Let $g_{\hbar,t}$ be the free whole-line evolution of (1), and let R reflect $q \mapsto -q$. Put

$$\gamma_\hbar = e^{-[m\omega q_0^2 + p_0^2/(m\omega)]/\hbar}, \quad \psi_{\hbar,D/N}(q,t) = \frac{g_{\hbar,t}(q) \mp g_{\hbar,t}(-q)}{\sqrt{1 \mp \gamma_\hbar}} \quad (q > 0). \tag{D0}$$

The upper sign is Dirichlet, the lower Neumann. Define the real half-line current form and full boundary stress

$$J_\hbar(t) = \hbar\,\mathrm{Im}\int_0^\infty \overline{\psi}\,\partial_q\psi\,dq, \quad F_\hbar(t) = \frac{\hbar^2}{2m}\left(|\psi'(0,t)|^2 - \mathrm{Re}[\overline{\psi(0,t)}\psi''(0,t)]\right). \tag{D1}$$

For both indicated wave families, prove $J_\hbar' = F_\hbar$ and, for every bounded continuous $\varphi : \mathbb{R} \to \mathbb{R}$,

$$\lim_{\hbar\downarrow 0}\int_{\mathbb{R}} \varphi(t) F_{\hbar,D/N}(t)\,dt = 2|p_0|\varphi(t_*). \tag{D2}$$

The Neumann measures are finite **signed** measures; their Dirichlet-only derivative term vanishes. At each fixed $s \in \mathbb{R}$ also prove

$$\lim_{\hbar\downarrow 0}\sqrt{\hbar}\,F_{\hbar,D/N}(t_* + \sqrt{\hbar}\,s) = \frac{2p_0^2}{m}\sqrt{\frac{m\omega}{\pi A(t_*)}}e^{-\Lambda^2 s^2}, \quad J_{\hbar,D/N}(t_* + \sqrt{\hbar}\,s) \to |p_0|\,\mathrm{erf}(\Lambda s). \tag{D3}$$

This is weak convergence of measures, not convergence in total variation to the delta. $J_\hbar$ is a current form: the half-line operator $-i\hbar\partial_q$ is not being asserted self-adjoint. No general Robin packet limit is part of this target.

7. Target E: two preparations and mixtures at a C^1 cusp

In units $m = 1$, fix $R > 0$ and a smooth even cutoff χ equal to one on $[-R, R]$ and zero outside $[-2R, 2R]$. Set

$$V(q) = -\tfrac{2}{3}|q|^{3/2}\chi(q), \quad F = -V', \quad H_\hbar = -\tfrac{\hbar^2}{2}\partial_q^2 + V(Q), \quad F_* = \|F\|_\infty.$$

Choose $T > 0$ such that $F_* T^2/2 < R/4$. Choose $C_0 > 0$ with $|F(x)| \le C_0\sqrt{|x|}$ for all x, and a fixed $K > 0$ such that $K^2 > \max\{16\sqrt{2}C_0, 16/3\}$. For $0 < \hbar < e^{-1}$ put $a_\hbar = 1/\log(1/\hbar)$ and

$$g_{\hbar,k}(q) = (\pi\hbar)^{-1/4}\exp\left[-\frac{(q-a_\hbar)^2}{2\hbar} - \frac{ika_\hbar^{3/4}(q-a_\hbar)}{\hbar}\right], \quad k \in \{0, K\}. \tag{E0}$$

These two states have exactly the same initial position density. Let $\psi_{\hbar,k}(t) = e^{-itH_\hbar/\hbar}g_{\hbar,k}$ and set

$$z_+(t) = (t^4/144, t^3/36), \qquad z_\times(t) = \begin{cases} z_+(t), & t \le 0, \\ -z_+(t), & t \ge 0. \end{cases} \tag{E1}$$

The target is uniform on $[-T, T]$:

$$(\langle Q\rangle_{\hbar,0}, \langle P_\hbar\rangle_{\hbar,0}) \to z_+, \quad (\langle Q\rangle_{\hbar,K}, \langle P_\hbar\rangle_{\hbar,K}) \to z_\times, \quad \mathrm{sd}\,Q_{\hbar,k} + \mathrm{sd}\,P_{\hbar,k} \to 0. \tag{E2}$$

For every $\lambda \in [0,1]$ and $\theta \in \mathbb{R}$, normalize $\sqrt{\lambda}\,g_{\hbar,0} + e^{i\theta}\sqrt{1-\lambda}\,g_{\hbar,K}$. For every $b \in C_c^\infty(\mathbb{R}^2)$, its evolved Weyl expectation must converge uniformly on $[-T, T]$ to

$$\lambda b(z_+(t)) + (1-\lambda)b(z_\times(t)). \tag{E3}$$

Thus both limiting incoming histories agree for $t \le 0$ and can have different outgoing measures. The target neither asserts finite-$\hbar$ Wigner positivity nor classifies waiting-time branches.

8. Target F: a relativistic joint limit

Let $V \in C^\infty(\mathbb{R};\mathbb{R})$ be bounded below with all derivatives of order at least two bounded, $m, c_0, h_0, T > 0$, and $z_0 \in \mathbb{R}^2$. For $c \geq c_0$, define

$$T_c(p) = \sqrt{m^2c^4 + c^2p^2} - mc^2, \quad S_{\hbar,c,s} = e^{-isT_c(P_\hbar)/\hbar} e^{-isV(Q)/\hbar}. \tag{F0}$$

For the coherent packet (1), its piecewise-linear interpolated means, and the nonrelativistic Newton flow $\Phi_t(z_0)$, prove that there exists $C_T < \infty$, independent of $\hbar, c, \pi$, such that

$$\sup_{0\leq t\leq T} \|z_{\hbar,\pi,c}(t) - \Phi_t(z_0)\|_\omega \leq C_T(\hbar + |\pi| + c^{-2}) \tag{F1}$$

for all $0 < \hbar \leq h_0$, $c \geq c_0$ and positive partitions of $[0,T]$. At fixed $\hbar$, the self-adjoint $T_c(P_\hbar) + V(Q)$ converges in strong resolvent sense to $P_\hbar^2/(2m) + V(Q)$ as $c \to \infty$, and its unitary groups converge strongly, uniformly on compact time intervals.

The separate free wall target uses the *spectral* Dirichlet Hamiltonian $T_c(\sqrt{-\hbar^2\Delta_D})$, not the generally different killed nonlocal process. For the odd packet (D0), its position expectation satisfies, for $|t| \leq T$,

$$\left|\langle Q\rangle^D_{\hbar,c}(t) - |q_0 + p_0 t/m|\right| \leq \sqrt{\frac{2}{1-\gamma_\hbar}} \left[\sqrt{\frac{\hbar}{2m\omega}} + \frac{T}{m}\sqrt{\frac{\hbar m\omega}{2}}\right] + \frac{T|p_0|^3}{2m^3c^2}. \tag{F2}$$

The positive-energy one-particle operator in an external potential is not a construction of a Lorentz-covariant interacting field theory.

9. Target G: an exact normalized Euclidean tower

Let $V : \mathbb{R} \to [0,\infty)$ be continuous, $m, \hbar, T > 0$, and $x, y \in \mathbb{R}$. Let $\mathbb{P}_0$ be the Brownian bridge of variance parameter $\hbar/m$ from x to y on $[0,T]$. Define

$$W = \int_0^T V(B_t)\,dt, \quad Z = \mathbb{E}_0 e^{-W/\hbar}, \quad \rho = Z^{-1}e^{-W/\hbar}, \quad \rho_n = \mathbb{E}_0[\rho \mid \mathcal{F}_n], \tag{G0}$$

where $\mathcal{F}_n$ is generated by dyadic evaluations of B. The target is $Z > 0$ and, almost surely,

$$\mathbb{E}_0\rho_n = 1, \quad 0 < \rho_n \leq Z^{-1}, \quad \rho_n = \mathbb{E}_0[\rho_{n+1} \mid \mathcal{F}_n], \quad \rho_n \to \rho \text{ a.s. and in } L^1(\mathbb{P}_0). \tag{G1}$$

For arbitrary positive partitions with mesh tending to zero, the normalized densities obtained from the trapezoidal sums of $V(B_t)$ also converge to ρ in $L^1(\mathbb{P}_0)$; their polygonal pushforwards converge weakly on $C([0,T])$. The pushforward measures are not claimed to converge in total variation. The ρ_n are conditional expectations of the continuum density; they are not identified with the original bare finite-lattice density.

10. The bridge to newtonlean

The inspected newtonlean master commit is 8138e944. Its toolchain is Lean 4.34.1 core/Std only, with no mathlib. The current session did not compile it. Its FiniteEstimates.cell uses **drift then kick**:

$$\mathrm{DK}_s(q,v) = (q + sv,\ v + sa(q + sv)), \tag{H0}$$

whereas (2), after $v = p/m$ and $a = -V'/m$, is **kick then drift**:

$$\mathrm{KD}_s(q,v) = (q + s[v + sa(q)],\ v + sa(q)). \tag{H1}$$

Equality of the two cells is false. The following finite rational lemma is an exact interface target, requiring no quantum foundations.

Target H. Let $a : \mathbb{Q} \to \mathbb{Q}$ be Λ-Lipschitz, with $\Lambda \in \mathbb{Q}_{\geq 0}$. Let $s_0, \ldots, s_{n-1} \in \mathbb{Q}_{\geq 0}$, $\tau \in \mathbb{Q}_{>0}$, and put $\|(q, v)\|_\tau = |q| + \tau|v|$. Iterate KD and DK from the same rational state. Suppose the DK states $(\bar{q}_j, \bar{v}_j)$ satisfy $|a(\bar{q}_j)| \leq A$ and $|\bar{v}_j| \leq B$ for $j < n$, with $A, B \in \mathbb{Q}_{\geq 0}$. Then

$$\|z_n^{KD} - z_n^{DK}\|_\tau \leq (A + \tau\Lambda B) \sum_{j=0}^{n-1} s_j^2 \prod_{k=j+1}^{n-1} (1 + s_k/\tau)(1 + \tau\Lambda s_k). \tag{H2}$$

Indeed one cell differs by at most $(A + \tau\Lambda B)s_j^2$, and KD amplifies a previous discrepancy by at most $(1 + s_j/\tau)(1 + \tau\Lambda s_j)$. The empty product is one. The analogous statement for real data with $\sum_j s_j = T$ implies

$$\|z_n^{KD} - z_n^{DK}\|_\tau \leq (A + \tau\Lambda B)T|\pi|e^{(1/\tau + \tau\Lambda)T}. \tag{H3}$$

The rational statement is intended as a finite algebraic lemma; (H3) also uses real exponential bounds and is a separate completion step.

The existing GeneralForcePolygonCurve.polygonMap_uniform_convergence constructs a uniformly convergent position polygon on its own binary time and Cauchy-quotient representations, under a regional CentralOracle, force-sampling, confinement and short-window conditions. It is not already the general one-dimensional global $C^{1,1}$ result in Target A: arbitrary partitions, external real-time identification and general noncentral forces remain separate. To connect the models one must additionally instantiate a compatible sample oracle, identify completed coordinates and velocities with real $q, p/m$, prove that (H2)'s bounds hold, and show the two limits agree. The axis-embedded harmonic oscillator is a suitable special case; general V need not define an inward central force.

The Reverse/Classical/ClassicalTrajectory.lean interfaces assume localized trajectories, free dynamics, the momentum–velocity relation and additive impulses; the QuantumTheory carrier has no Hilbert-space or operator structure. Targets A–F could provide evidence for restricted, explicitly embedded versions of some of these interfaces. They do not automatically discharge the interfaces for arbitrary composite bodies or prove a quantum gravity parent. Historical Newton propositions keep their given-trajectory premise; none of these modern targets is a historical premise.

For context only, current mathlib documentation has Picard–Lindelöf, Schwartz Fourier analysis, and Brownian-motion definitions. Those pages are not a claim that this project has, or may import, the needed version or all operator-theoretic results. Under the present core-only rule, a full quantum development must construct its analytic foundations or remain explicitly conditional; a dependency expansion requires separate user direction. In either route, check the actual compiled theorem types and axiom dependencies at the pinned toolchain.

11. Ordering at finite mesh and in the joint limit

Result, written derivation, 2026-10-28; no priority claim. The ordering difference has three distinct levels: the two finite maps are different; either order, even mixed cell by cell, reaches the same *specified* quantum Hamiltonian when the mesh goes to zero at fixed $\hbar$; yet their quantum vectors need not approach the same ray along every joint $\hbar$, mesh limit. The common classical limit of the two schemes was already established in the ODE lesson, including mixed orders.

11.1 Fixed action and finite products

For each $\hbar > 0$, use the quantum operators of §2. With $V \in C^{1,1}$ and V' globally Lipschitz, allow **either** $D_s K_s$ or $K_s D_s$ in each cell. For every fixed $\hbar, T > 0$ there is a finite $C_{\hbar,T}$ such that every such mixed product obeys

$$\|S_{\hbar,\pi}^{\text{mixed}}(T)f - U_\hbar(T)f\|_2 \leq C_{\hbar,T}|\pi|\,\|N_\hbar f\|_2, \quad f \in D(N_\hbar). \tag{I1}$$

The proof swaps the kinetic and potential factors in the lift, equations (11a)–(13). The commutator changes sign but not norm; the same graph-growth exponents bound both orders. Apply either local bound on each cell and telescope along the exact orbit. Density and unitarity extend strong convergence to all L^2 states. Neither (I1) nor its constant asserts uniformity as $\hbar \to 0$.

At finite s the products generally differ. For a smooth subquadratic V and $f \in \mathcal{S}$, their second-order strong expansion is

$$(D_s K_s - K_s D_s) f = -\frac{s^2}{\hbar^2}[T_\hbar, V(Q)]f + o(s^2) = \frac{is^2}{2m\hbar}(P_\hbar V'(Q) + V'(Q)P_\hbar)f + o(s^2). \tag{I1a}$$

The remainder is for a fixed $\hbar$ and vector; no operator-norm Baker–Campbell–Hausdorff expansion for these unbounded operators is asserted.

For the coherent Gaussian (1), the spread and comparison estimates (A2) also hold with the corresponding mixed classical polygon. In a drift-first cell, apply the kick after the drift: the force deviation is bounded by L times the position spread *after* that drift, at most $(1 + s\omega)W_j$. This gives the same upper recurrence as in the lift:

$$\begin{aligned} W_{j+1} &\le (1 + s_j\omega)\left(1 + \frac{s_j L}{m\omega}\right) W_j, \\ e_{j+1} &\le (1 + s_j\omega)\left(1 + \frac{s_j L}{m\omega}\right) e_j + \frac{s_j L}{m\omega}(1 + s_j\omega)W_j. \end{aligned} \tag{I2}$$

Here e_j is the weighted distance between quantum means and the classical vertices using the **same** cell orders. Thus the bounds (A2) are valid for every mixed order. The ODE lesson gives the independent $O(|\pi|)$ classical polygon error. Both time-order schemes therefore recover the same classical means and smooth Weyl observable values for every joint $\hbar, |\pi| \to 0$.

For a uniform step s the finite classical maps are conjugate:

$$\mathrm{DK}_s = K_s^{\mathrm{cl}}\,\mathrm{KD}_s (K_s^{\mathrm{cl}})^{-1}, \qquad (\mathrm{DK}_s)^n = K_s^{\mathrm{cl}} (\mathrm{KD}_s)^n (K_s^{\mathrm{cl}})^{-1}. \tag{I3}$$

Exactly the same algebra holds for the quantum unitaries: $(K_s D_s)^n = K_s (D_s K_s)^n K_s^{-1}$. This is not equality on the same initial state: the conjugacy changes the endpoint identification. It gives no single conjugator for general unequal steps. The two first-order symplectic Euler choices and their adjoint relation are established in the variational-integrator literature (symplectic Euler example read).

11.2 Exact distinguishability under a constant force

Fix $m, \omega, T > 0$, $F \neq 0$, $q_0, p_0 \in \mathbb{R}$ and $V(q) = Fq$. On $\mathcal{H} = L^2(\mathbb{R})$, start both products from the same Gaussian (1). For any positive partition write

$$A_\pi = \sum_j s_j^2, \quad \Delta q_\pi = \frac{F}{m} A_\pi, \quad C_\pi = \frac{F^2}{2m} \sum_j s_j^2 (2T - t_j - t_{j+1}). \tag{I4}$$

Let ψ_{KD} use kick first in every cell and ψ_{DK} use drift first in every cell. With the $\hbar$-scaled momentum transform $\widehat{\psi}(p) = (2\pi\hbar)^{-1/2} \int e^{-ipq/\hbar}\psi(q)dq$, the **exact** relation is

$$\widehat{\psi}_{DK}(p) = e^{-i(C_\pi + \Delta q_\pi p)/\hbar} \widehat{\psi}_{KD}(p). \tag{I5}$$

Consequently

$$|\langle \psi_{KD}, \psi_{DK} \rangle| = \exp\left[-\frac{\omega F^2}{4m\hbar} A_\pi^2\right], \quad \inf_{\theta \in \mathbb{R}} \|\psi_{KD} - e^{i\theta}\psi_{DK}\|_2^2 = 2\Big(1 - e^{-\omega F^2 A_\pi^2/(4m\hbar)}\Big). \tag{I6}$$

Indeed, in momentum space a kick sends $\widehat{\psi}(p)$ to $\widehat{\psi}(p + Fs)$ and a drift multiplies it by $e^{-isp^2/(2m\hbar)}$. At cell j, the drift momentum in the kick-first product is $p + F(T - t_{j+1})$ and in the drift-first

product it is $p+F(T-t_j)$. Subtracting the accumulated quadratic phases gives (I5). Both final momentum densities are Gaussian with variance $\hbar m\omega/2$; their characteristic function gives (I6).

For the uniform partition $s=T/n$, take $\hbar_n=\kappa s^2$ with any fixed $\kappa>0$. Then $A_\pi=Ts$ and

$$\inf_\theta \|\psi_{KD}-e^{i\theta}\psi_{DK}\|_2^2 = 2\Big(1-e^{-\omega F^2T^2/(4m\kappa)}\Big) > 0 \quad \text{for every } n. \tag{I7}$$

Nevertheless, both momentum means are p_0-FT; their position means are $q_{\mathrm{cl}}(T)\mp FA_\pi/(2m)$, and their position and momentum standard deviations are $O(\sqrt{\hbar})$ on $[0,T]$. Thus both families converge to the **same** classical point for every joint schedule, while their rays can remain separated, or even become orthogonal when $A_\pi^2/\hbar\to\infty$.

The distinction between semiclassically uniform observable accuracy and wavefunction accuracy is itself established; see Fang–Qu (abstract read). The calculation here compares the two specified first-order products directly and gives their exact overlap.

For $F\neq 0$, (I6) says ray distance tends to zero exactly when $A_\pi/\sqrt{\hbar}\to 0$. A ray test ignores the scalar factor in (I5): convergence of rays does not by itself control relative phase or interference. At fixed $\hbar$, $A_\pi\le T|\pi|\to 0$, and the phase in (I5) also vanishes on the relevant state. This is a real finite-resolution ordering effect, not a different fixed-action continuum Hamiltonian. It complements the polygon phase note, which treats different histories with matching classical endpoints and a residual scalar phase. The present endpoints differ at finite mesh. No literature priority is claimed for (I5)–(I7).

11.3 Operator ordering is a further question

The previous result assumes the **same** operator $H_\hbar=P_\hbar^2/(2m)+V(Q)$. It does not settle factor-ordering ambiguities when a classical Hamiltonian contains mixed functions of position and momentum. For a concrete target, take $f(q)=2+\cos q$, and on Schwartz space define

$$H_\hbar^{(1)}=\frac{1}{2m}P_\hbar f(Q)P_\hbar, \qquad H_\hbar^{(2)}=\frac{1}{4m}\big(f(Q)P_\hbar^2+P_\hbar^2 f(Q)\big). \tag{I8}$$

Both are symmetric and have the same classical symbol $f(q)p^2/(2m)$, but the exact operator identity on Schwartz space is

$$H_\hbar^{(1)}-H_\hbar^{(2)}=\frac{\hbar^2}{4m}f''(Q)=-\frac{\hbar^2}{4m}\cos Q. \tag{I9}$$

The first operator has its standard nonnegative uniformly elliptic self-adjoint realization on $H^2(\mathbb{R})$; the second is its bounded self-adjoint perturbation by $+\hbar^2\cos Q/(4m)$. For any Schwartz initial state the difference of their initial momentum derivatives is $-\hbar^2\langle\sin Q\rangle/(4m)$, which need not vanish. Thus a common classical symbol alone does not determine a unique finite-$\hbar$ dynamics. In more general path integrals, time-slicing and factor ordering can interact: see Dewey (1990) and Bastianelli–Schalm–van Nieuwenhuizen (1998) (abstracts read). No general uniqueness of the quantum lift follows.

Targets (I1)–(I9) are written proof/verification targets, not compiled Lean theorems. Formalizing (I5)–(I7) also requires the scaled Fourier transform, Gaussian normalization and characteristic function, and the phase-space/ray norm conventions to be defined rather than assumed as the conclusion.

12. Consequence for STATE

These targets distinguish standard inputs, repository-specific statements and unknown publication priority. They expose a finite classical comparison that can be attempted under newton-lean's present rules, and the substantial analytic foundations needed for an end-to-end quantum verification. The constant-force calculation separates fixed-action slice independence from joint-limit distinguishability and from operator-ordering ambiguity. The universal lifting conjecture, positive-action necessity and the four-dimensional gauge construction remain open.